

Bilateral nonarteritic ischemic optic neuropathy and retinal ischemia in a pediatric dialysis patient

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A nearly 3-year-old boy on nightly dialysis presented emergently with sudden loss of vision. On examination, his visual acuity was light perception in the right eye and no light perception in the left eye. There was bilateral optic disk edema, diffuse pallor of posterior poles, and a cherry red spot in the left fundus. The patient was subsequently found to be hemodynamically unstable and admitted to the pediatric intensive care unit with presumed septic shock. Optical coherence tomography revealed paracentral acute middle maculopathy lesions in the right eye and diffusely thick retina in the left eye. Magnetic resonance imaging and magnetic resonance angiography of the brain and vessels did not reveal any acute findings. The patient's presentation was most consistent with bilateral nonarteritic ischemic optic neuropathy and unilateral central retinal artery occlusion. On repeat evaluation 9 months later, vision was largely unchanged.

Case Report

A boy aged 2 years and 11 months with no prior ocular history was in his usual state of health when he presented emergently at Ann & Robert H. Lurie Children's Hospital of Chicago with sudden, painless binocular vision loss. The patient had end-stage renal disease (ESRD) on nightly peritoneal dialysis (PD) secondary to congenital Finnish-type nephrotic syndrome status post bilateral nephrectomy at 10 months of age.

Visual acuity on presentation was light perception in the right eye and no light perception in the left eye. Pupils were dilated and minimally reactive. There was bilateral optic disk edema, with absent cups and tortuous vessels. The right

eye had multiple patchy confluent areas of pale retina throughout the posterior pole, while the left fundus had diffuse pallor with a cherry-red spot. The patient was alert and responsive but found to be hypotensive (blood pressure in the 60s/40s) and tachycardic (150-160 beats per minute). He was admitted to the pediatric intensive care unit (PICU), where he was fluid resuscitated and started on vasopressors plus broad-spectrum antibiotics for presumed decompensated septic shock in the setting of leukocytosis and fever. Blood, peritoneal fluid, and cerebrospinal fluid cultures were obtained, which were negative for growth during admission.

Eye examination was performed at the same time as imaging under anesthesia less than 24 hours after initial presentation and once the patient was hemodynamically stable. Color fundus photographs (Figure 1A), red-free fundus photos (Figure 1B), fluorescein angiography (FA), and optical coherence tomography (OCT) images were obtained. On OCT there were hyper-reflective bands in the inner nuclear layer of the right eye, consistent with paracentral acute middle maculopathy (PAMM) (Figure 1C); in the left eye, there was diffusely thickened inner retina (Figure 1D). FA showed poor choroidal flush (likely secondary to blockage from macular edema) with normal arm-to-retina transit (16 sec) and arteriovenous (22 sec) filling times. There was late leakage from both optic nerve heads as well as a few patchy areas of peripheral vascular branching.

Magnetic resonance imaging and magnetic resonance angiography of the brain and vessels were obtained to look for acute ischemic injury elsewhere in the central nervous system. These studies showed no evidence of any acute diffusion restriction or angiographic anomaly; however, they did show prominent anterior optic nerve heads as well as areas of periventricular gliosis, likely reflecting old ischemia.

At 5 weeks' follow-up, there was minimal improvement in vision, with bare light perception in the right eye and no light perception in the left eye. Optic disk edema resolved with consequent pallor (left greater than right) (Figure 1E). Optic disk drusen were noted in both eyes (left greater than right) on indirect ophthalmoscopy and red-free fundus photographs. OCT imaging showed bilateral thinning of the inner retina without macular edema. Repeat FA findings were unremarkable. At 9 months after initial presentation, the patient's vision remained largely unchanged. Visual acuity testing was limited, secondary to the patient's developmental delay, but was at least light perception in the right eye, no light perception in the left eye, with ability to cautiously ambulate in the examination room. On serial examinations, there remained no evidence of neovascularization 2.5 years following initial presentation.

Discussion

Acute blindness due to nonarteritic anterior ischemic optic neuropathy (NAION) or central retinal artery occlusion (CRAO) are infrequent but known complications of hypotension in patients on chronic dialysis.¹⁻³ There have been previous reports of bilateral NAION specifically in children on chronic PD or intermittent hemodialysis.⁴⁻⁷

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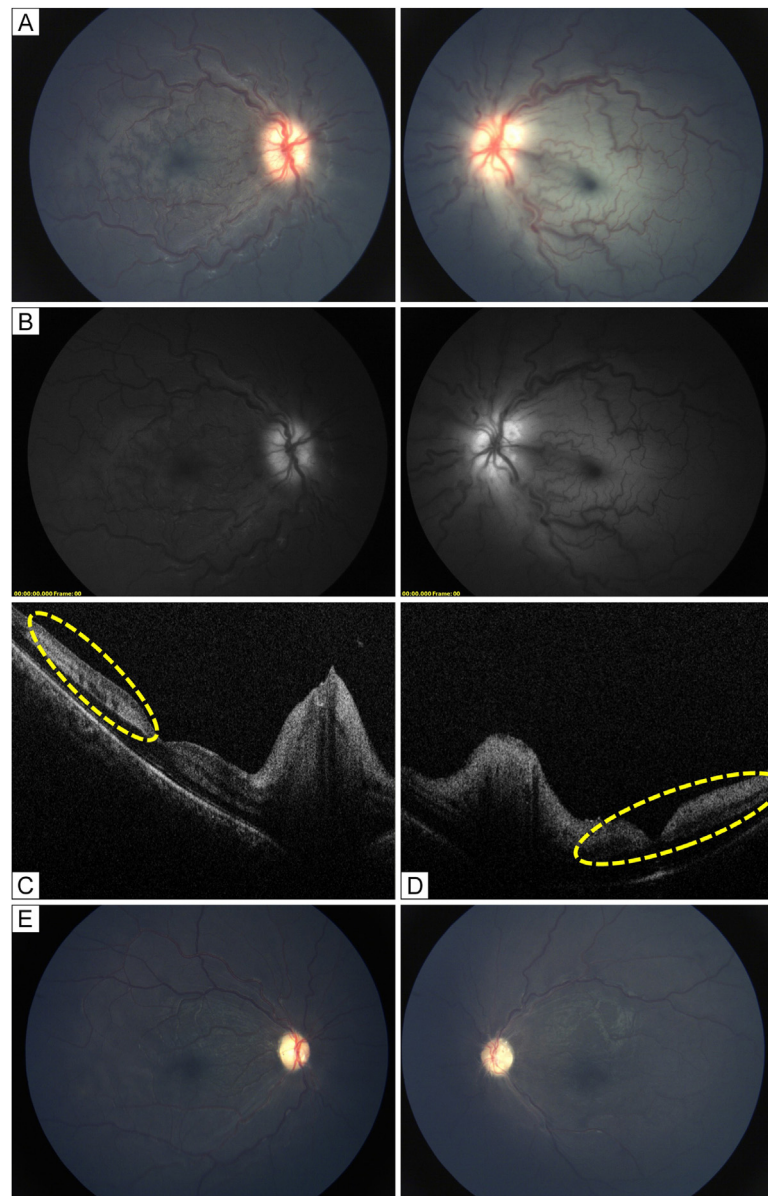


FIG 1. Multimodal imaging. A, Color fundus photography at presentation demonstrating bilateral swollen optic disks, bilateral pale macula, and cherry red spot in the left fundus. B, Red-free photographs at initial presentation highlighting the tortuosity of the retinal vasculature. C-D, Optical coherence tomography of the right (C) and left eyes (D) showing hyper-reflective paracentral acute middle maculopathy (PAMM) lesions in the right eye and thickened, hyper-reflective inner retina in the left eye. E, Fundus photograph demonstrating resolution of optic disk swelling with subsequent pallor at 5 weeks.

An Italian study in 2015 proposed that the incidence of NAION in children undergoing chronic dialysis was approximately 1%.⁶ A key contributing factor to optic nerve and retinal hypoperfusion in PD patients is acute and chronic hypotension as well as elevated blood viscosity and anemia.² The main causes of hypotension in this setting are thought to be chronic and acute hypovolemia, antihypertensive therapy, anephric status, and myocardial failure in addition to other poorly understood causes, such as autonomic neuropathy and adrenal insufficiency.⁶

The proposed pathophysiology of sudden vision loss due to hypotension in these patients is related to two opposing

forces. During dialysis, there is a theoretical increase in intraocular pressure caused by the rapid removal of urea causing an osmotic shift of water from the plasma into the aqueous fluid.⁸ Chronic dialysis may predispose patients to increased risk of ischemic insult in the setting of dialysis-related hypotension compounded by an element of nocturnal hypotension. The combination of elevated intraocular pressure and hypotension could thus create a pressure gradient that could precipitate hypoperfusion of the posterior ciliary vessels or retinal circulation leading to NAION or CRAO, respectively.² Although a normal eye might be able to compensate with dilation of vascular

beds, patients with damaged autoregulatory mechanisms, which is common in uremic patients,⁹ may be unable to respond. ESRD patients are also at risk of hypercoagulable states, thought to be secondary to dysregulation of the coagulation cascade, platelets, and vessel wall in the setting of uremic toxins and accumulation of metabolic compounds.¹⁰ This is a known risk factor for development of NAION¹⁰ or CRAO,¹¹ especially in young patients. Of note, like other cases in the literature,⁵⁻⁷ our patient did not have signs of acute ischemia on brain imaging. This may be explained by shunting of blood away from the eye to favor the brain in the setting of hypotensive crisis. The areas of gliosis in our patient may be suggestive of hypoperfusion from chronic hypotension.⁵

Unlike other patients previously reported, ours uniquely had bilateral NAION in addition to a unilateral CRAO and OCT finding of PAMM in the fellow eye, indicating infarction of the retrolaminar portion of the optic nerve head supplied by the short posterior ciliary arteries, inner retina supplied by the central retinal artery, and presumably inner nuclear layer infarction supplied by the retinal capillary plexus,¹² respectively. Our patient was promptly admitted to the PICU for presumed septic shock at the time of presentation, so it is possible that septic shock in addition to the fluctuating hemodynamics in the setting of PD contributed to the devastating optic nerve and retinal ischemic insult that occurred in our patient. Our patient was also found to have optic disk drusen, which may have increased his predisposition to NAION and CRAO. This case highlights the extent of ischemic injury that can take place in the setting of hypotension, specifically in a pediatric dialysis patient.

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